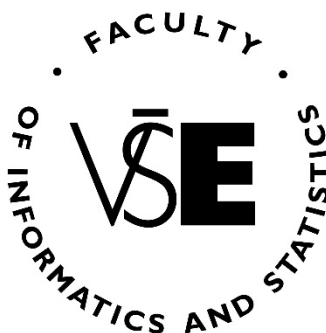


Prague University of Economics and Business

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**DEVELOPMENT OF A PROGRAM FOR
OPTIMIZING A SECURITIES PORTFOLIO ON
THE RUSSIAN STOCK MARKET**

BACHELOR THESIS

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Abstract

Effective portfolio management is paramount for investors seeking to achieve optimal returns while managing risks in the dynamic environment of the Russian stock market. This thesis addresses to develop a Python program for identifying the optimal portfolio of Russian stocks using two widely known models: Mean-Variance and Mean-Semivariance model.

The research is divided into two main parts. In the initial phase, we delve into the modern portfolio theory, identifying key parameters and methodologies for portfolio optimization. This includes an exploration of risk-return trade-offs, performance evaluation metrics, and optimization techniques. Subsequently, the focus shifts to the practical implementation of the theoretical concepts discussed. We outline the process of retrieving real-time data using API and leveraging Python programming to formulate and solve the mathematical model for portfolio optimization. Through empirical analysis and comparative studies, we evaluate the effectiveness of the developed Python program in constructing optimized portfolios tailored to the Russian stock market. In the concluding section, we synthesize the findings, highlight the practical implications of the research, and provide recommendations for investors seeking to leverage technology-driven solutions for portfolio management.

Keywords

Efficient Frontier, Markowitz model, Mean-Semivariance model, Mean-Variance model, Russian stocks.

Abstrakt

Efektivní správa portfolia je prvořadá pro investory, kteří se snaží dosáhnout optimálních výnosů při řízení rizik v dynamickém prostředí ruského akciového trhu. Tato práce se zabývá vývojem Python programu pro identifikaci optimálního portfolia ruských akcií pomocí dvou široce známých modelů: Mean-Variance a Mean-Semivariance model.

Výzkum je rozdělen do dvou hlavních částí. V počáteční fázi se ponoříme do moderní teorie portfolia, identifikujeme klíčové parametry a metodiky pro optimalizaci portfolia. To zahrnuje zkoumání kompromisů rizika a návratnosti, metriky hodnocení výkonu a optimalizační techniky. Následně se zaměření přesouvá na praktickou implementaci diskutovaných teoretických konceptů. Nastíníme proces načítání dat v reálném čase pomocí API a využití programování v Pythonu k formulování a řešení matematického modelu pro optimalizaci portfolia. Prostřednictvím empirické analýzy a srovnávacích studií hodnotíme účinnost vyvinutého Python programu při vytváření optimalizovaných portfolií přizpůsobených ruskému akciovému trhu. V závěrečné části syntetizujeme zjištění, zdůrazňujeme praktické důsledky výzkumu a poskytujeme doporučení pro investory, kteří se snaží využít technologicky řízená řešení pro správu portfolia.

Klíčová slova

Efektivní Hranice, Markowitzův model, Mean-Semivariance model, Mean-Variance model, Ruské akcie.

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Introduction

Investing in the market through a portfolio approach is widely regarded as the most effective strategy for optimizing returns while managing risk. This method is applicable to both individual investors and institutional ones. Conceptually, an investment portfolio can be likened to a pie, with each slice representing an asset with different parameters. Through strategic allocation of assets, investors aim to achieve a balanced portfolio that optimize risk and return. While various securities can contribute to portfolio diversification, this paper will specifically focus on equities as a key component of investment portfolios.

The choice of this topic stems from my professional experience as a financial analyst working on a media project in Russia dedicated to individual investors. Through my work in this role, I became increasingly fascinated by the intricacies of portfolio management. It gave me valuable insights into how the market operates, how investors behave, and why prudent investment strategies matter. It was during this period that I developed a keen interest in exploring portfolio management methodologies and their application in real-world investment scenarios. This background inspired me to delve deeper into the subject and understand the nuances of portfolio optimization within the context of the Russian stock market. Moreover, many people are primarily interested in their domestic market, and considering that Russian citizens currently face restrictions on investing in international markets, I believe that my work can contribute to fostering a better understanding of sound investment practices in stocks for my fellow citizens in the future.

The main objective of this thesis is to develop a program for optimizing Russian stock portfolio using several concepts based on modern portfolio theory. This can help the private investor in deciding how best to allocate his available funds among possible investment opportunities. We will find and compare optimal portfolios of shares traded on the Moscow Exchange based on data collected over the last 10 years from 2013 to 2023. This will be accomplished by employing two mean-risk models: the Mean-Variance model and the Mean-Semivariance model. The choice of these models is particularly driven by their robustness in capturing the dynamics of risk and return which are critical in the volatile Russian market environment. The Mean-Variance model is renowned for its foundational approach in balancing risk against expected returns, while the Mean-Semivariance model is selected for its focus on downside risk, which is particularly relevant given the turbulent nature of the Russian financial market. A secondary goal of the work is to compare the results with a Moscow Exchange Index (MOEX).

In the first part of thesis, we will delve into the historical context of the Russian stock market, exploring its evolution and unique characteristics, problems, and trends in the development of local stock exchanges. The second part will focus on an in-depth analysis of portfolio theory, providing the theoretical foundation upon which the optimization program will be developed. This section will serve as the basis for understanding key concepts such as risk-return trade-offs and portfolio diversification. Finally, in the third part, we will embark on the practical implementation of our research. This will involve the collection and cleansing of data related to all traded stocks over the past decade. Subsequently, we will develop optimization models in Python and conduct a comprehensive comparative analysis to evaluate their effectiveness in constructing optimal portfolios tailored to the Russian stock market.

1 Stock Market

The stock market is the most important instrument of financing and development of the economy. It plays a special role as it is designed to transform savings of households and enterprises into investments. The formation of stock exchange infrastructure is a consequence of the development of market relations. The need for the emergence of this institution was due to the expansion of trade in a number of goods, such as commodities and agricultural products, and later securities. Buyers and sellers seek to maximize their profits from a transaction, so they want to be sure that the transaction price reflects the current supply and demand relationship. Therefore, the market for such commodities must bring together and summarize a large amount of information, and in the short term. To fulfil this function, it must be highly centralized. This goal is achieved in the organization of the stock exchange, where the supply and demand of goods collide. During the evolution of stock exchanges, new instruments emerged, the trading process was improved and modernized. All this led to the process of globalization and universalization of exchange activities, erasure of distinctions between different types of exchanges, development of processes of concentration and consolidation of exchange trading systems acquiring international character, elimination of intermediate links from the process of concluding transactions. The same applies to Russian exchanges.

The stock exchange is an organized, constantly functioning market where securities are bought and sold. It mobilizes funds for long-term investments in the economy and for financing government programs, buying and selling shares, bonds of joint-stock companies and government loan bonds (Lavrenova, 2007)

1.1 Russian stock market

Russian stock exchanges lag far behind foreign stock exchanges in terms of their development. The reason for this is not only the limited range of financial instruments in Russia, but also the lack of the necessary legislative framework. Poor infrastructure development, as well as the absence of a modern system for stock exchanges in the country restrain the market potential. Another reason for the slowdown in the development of domestic stock exchanges is Russia's current position in the international arena. Due to a number of sanctions imposed on Russia, the stock market has undergone significant negative changes.

Each country's stock market is influenced by the financial globalization and liberalization of other countries, while the level of international financial integration depends on the effectiveness of macroeconomic exchange rate policies (Central bank of the Russian Federation, 2022). Russia has a mixed model of the financial market, where both commercial banks, which have all rights to securities transactions, and non-bank investment institutions are present at the same time and with equal rights. The securities market in Russia is relatively young and dynamic with rapidly growing volumes of transactions.

In total, as of March 2024, there are such exchanges operating in Russia as:

- Moscow Exchange;
- Saint Petersburg Stock Exchange;

- Saint-Petersburg Currency Exchange;
- Siberian Interbank Currency Exchange;
- Nizhny Novgorod Stock Exchange (Afanasieva, 2021).

1.2 Moscow Exchange

The largest stock exchange in Russia is the Moscow Exchange, where all major Russian issuers are traded. It was formed through the merger of MICEX Group and RTS Group in December 2011, and in February 2013 it launched an IPO on its own trading floor (Ozherelkova, 2023). Moscow Exchange indices are key indicators of the Russian organized market of securities and futures contracts. Moscow Exchange calculates indexes of stocks, bonds, mixed indexes, as well as indicators of derivatives and foreign exchange markets. The main equity indexes are the Moscow Exchange Index (MOEX Index) and the Russian Trading System (RTS Index), which are calculated on the same basis of securities but in different currencies (the MOEX Index is calculated in rubles, the RTS Index in US dollars) (MOEX, 2024).

The stock market in Russia is quite young; for this reason, many Western rating agencies continue to classify it as an emerging market characterized by higher returns due to a higher degree of risk. Since the mid-2000s, the domestic stock market has witnessed a number of positive changes: an increase in the total volume of sales, improved market liquidity, increased information transparency of its participants, strengthening of the legislative framework, emergence of new and strengthening of the existing mechanisms for investor rights protection (Ozherelkova, 2023).

Today the stock market is one of the main elements of the country's financial system. It is influenced by the general trend of social and economic development of the country. Especially under the influence of negative factors of the market environment, observed since the beginning of the period of sanctions imposed in 2014. Among them we can mention the growth of inflation, ruble devaluation, decline in real GDP growth rates, fall in purchasing power, outflow of foreign direct investments (Tikhonov & Karabut, 2023).

Despite all the negative processes, the ruble devaluation, although acting as a serious factor for the tendency of the Russian stock market development, was not considered a significant obstacle for investors. This is confirmed by the MOEX stock index, the data of which are shown in Figure 1.

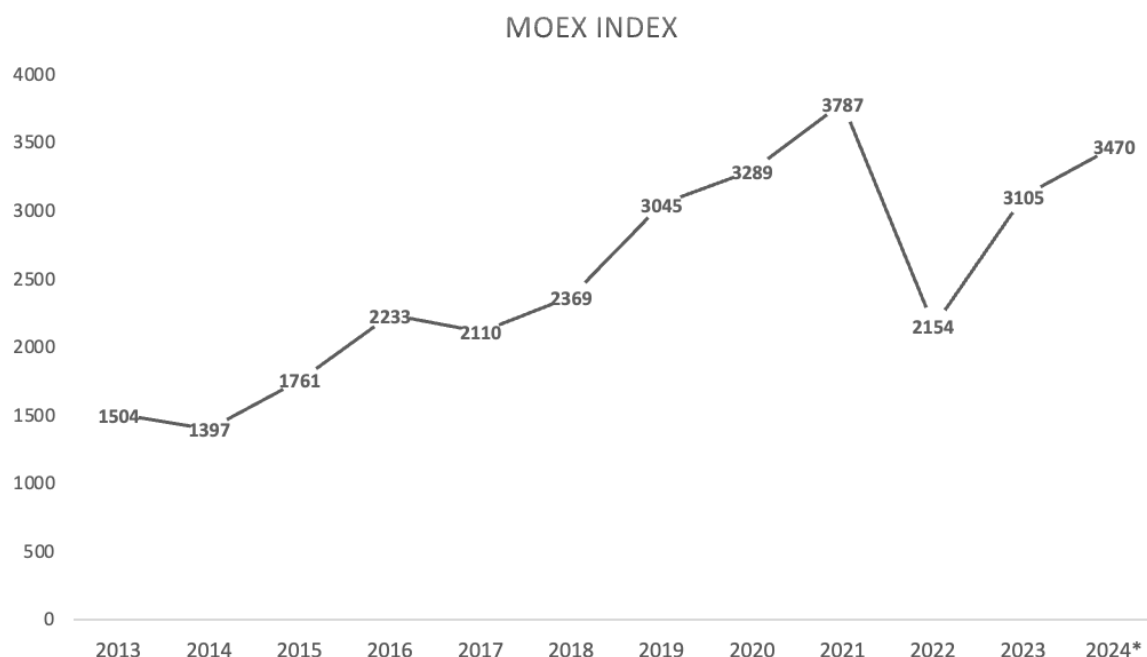


Figure 1 Moscow Stock Exchange Index 2013-2024 (TraidingView, 2024)

The data for 2024 is based on the trading session as of 1 May, with the remainder as of 31 December.

The MOEX exchange index grew from 1396 points in 2014 to 3787 in 2021. On 24 February 2022, there was an all-time record drop in the Moscow Exchange Index - by 45.4% - to 1684.95 points, and by the beginning of 2024 it grew to 3296.3 points. The top 10 companies by capitalization as of 1 May 2024 on MOEX are presented in Table 1.

Company name	Ticker	Market Cap
Sberbank	SBER	\$71,19 B
Rosneft	ROSN	\$66,00 B
Lukoil	LKOH	\$59,94 B
Gazprom	GAZP	\$41,34 B
Novatek	NVTK	\$40,13 B
Nornickel	GMKN	\$25,17 B
Polyus	PLZL	\$19,25 B
Severstal	CHMF	\$17,27 B
Tatneft	TATN	\$16,74 B
Novolipetsk Steel	NLMK	\$15,50 B

Table 1 Top 10 companies by capitalization on MOEX (Global Ranking, 2024)

1.3 Problems and trends in the development of stock exchanges in Russia

Of all the segments of the Russian financial market, it is the stock market that has been lagging behind in recent years. Moreover, this situation is very different from the mid-2000s, when the capitalization of the Russian stock market exceeded 100% of GDP and was 5 times larger than bank assets. In recent years, the share of the Russian stock market in the total value of Russian financial assets has been consistently declining. The capitalization of the Russian stock market at the end of 2021 was USD 754.7 billion. As of March 2024, the capitalization of the Russian stock market amounted to USD 650 billion, which is comparable to the Spanish stock market and represents 33% of GDP. It should be noted that in 2007 the capitalization of the Russian stock market was over USD 1 trillion. At the same time, 2022 brought previously unimaginable problems to the reality of the Russian stock market, which will continue to persist in the future (Igonina, 2009).

The market is more isolated and left to its own devices than ever before - there are no significant cross-border investment flows for the Russian equity market, nor are there likely to be in the foreseeable future. On the one hand, this means that, unlike other emerging markets, Russia's attractiveness does not matter in comparison to other markets at any given point in time (BCS Investments, 2022).

However, the absence of global institutional investors has led to a very unusual and counterintuitive trading regime. It is characterized by general low activity, volatile trading volumes and highly speculative stock movements typical for private investors. Finally, if Russian private investors are able to generate substantial income, they will come to the conclusion that the stock market remains one of the few available instruments to preserve savings and protect against inflation. However, for such a potential positive effect it is necessary that the real disposable income of the population grows steadily.

1.4 Trends in the development of stock exchanges in Russia

For the transition of the Russian market to the next stage of development, as well as for its integration into the international investment process, it is necessary to apply modern forms of investment activities that are successfully applied in countries with developed stock markets. This is achieved by converting the economy's savings into investments, and can also serve as a source of investments for domestic production by international financial markets.

Sustainable long-term growth of the Russian economy can be achieved through large-scale internal and external investments. At present, there is an acute shortage of funds in most sectors of the Russian economy, with the exception of export-oriented industries. The volume of investments in Russia is insufficient to speak about the country's significant participation in international investment flows (Central bank of the Russian Federation, 2022).

Given the scale of the economy, it can be said that the Russian stock market has significant growth potential. In the future, it may serve as a solution to the key problems of the Russian market, namely modern corporate development, and modernization of national production. At the moment, the stock market in Russia is rather a way of redistributing property than a mechanism of economic development.

2 Portfolio theory

This chapter explores the theoretical foundations of modern portfolio theory, with a specific focus on elucidating the mathematical underpinnings of two pivotal models: the Mean-Variance and the Mean-Semivariance. The discourse will extend to a thorough examination of the methodologies employed in calculating these models, underscoring their significance in the landscape of financial analytics. Additionally, the chapter will explore metrics used to evaluate the optimality of the portfolios derived from these models.

2.1 Traditional portfolio theories

Before the advent of Harry Markowitz's modern portfolio theory, investment approaches and portfolio management were often based on less rigorous and less mathematical techniques.

The following portfolio theories are distinguished based on their categorization by time of emergence (Rutterford & Sotiropoulos, 2016).

- Dow Theory. Focused on analyzing trends and cyclicalities in asset prices, applying technical analysis to generate buy or sell signals. This theory did not include formal portfolio risk management.
- Random walk theory. This theory, and the efficient market theory close to it, describes asset price movements as unpredictable and based on random events, emphasizing the difficulty of outperforming the market through stock selection and within the present discourse does not provide any analytical tools in terms of portfolio management. Although random walk theory mentions risk, saying that it is impossible to consistently generate above-market risk-adjusted returns, but the concept of risk itself is not analytically defined. This theory formed the basis for understanding the importance of diversification and supported the idea of passive index investing.
- Early formal approaches. Some proposed formula-based asset allocation rules, seeking to minimize potential losses, but without a modern understanding of the relationship between risk and return. Rather, these approaches sought ways to manage investments through a "naive" understanding of risk based on intuitive judgements.

2.2 Modern portfolio theory

Only in 1952, Markowitz G. (Markowitz, 1952) proposed a formal approach to the definition of risk, showed the functional dependence of profitability on risk and proposed a formal methodology of portfolio selection according to the risk/return ratio. This theory is now known as "Modern Portfolio Theory", which gave rise to the application of analytical methods in the evaluation of portfolios of financial assets.

Modern portfolio theory states that every investor seeks to realize two goals: to maximize returns and to minimize the risk associated with the fact of receiving such returns.

Thus, at the moment of making a decision to purchase a security or a set of securities, it is necessary to balance these two objectives, which is what the proposed theory helps to do.

The basic tenets of the theory are as follows:

At the moment t_0 the investor does not know what the level of returns of potential portfolios will be, so the expected returns should be considered as a random variable. A random variable, among other things, has two characteristics - mean value (mathematical expectation) and standard deviation. Therefore, an investor needs to evaluate the expected returns of potential portfolios, which in the vast majority of cases are expressed as average historical returns over a certain holding period and standard deviations or risks, which are also in many cases expressed as standard deviation of historical returns over a certain holding period. And then select the best portfolio in terms of maximizing returns at the lowest acceptable risk.

It is worth noting that this theory implicitly makes two assumptions:

1. For the same level of uncertainty, an investor will always prefer a higher yield.
2. For the same return, an investor will always prefer minimum risk.

2.2.1 Rate of return

Consider a portfolio with a quantity n representing the count of different assets within it. Let's denote m as the frequency of data points available, which could, for example, equate to the number of days being considered. Define p as the price series for a given asset at a specific point in time t , and let r symbolize the series of returns at time t . There are two distinct methods to compute returns:

1) Linear Method

$$r_t = \frac{P_{t+1} - P_t}{P_t} \quad (1)$$

where r_t is the rate of return at time t , calculated arithmetically;

$t = 1, 2, \dots, T$, where T is the number of time periods.

2) Logarithmic Method

$$r_t = \ln\left(\frac{P_{t+1}}{P_t}\right) \quad (2)$$

where r_t is determined using the natural logarithm to represent the rate of return.

The exponential, or logarithmic, method is often preferred for historical financial analysis for a couple of reasons. Firstly, it simplifies the process of aggregating periodic returns to find the total over an extended period, like a year. This is because the logarithmic returns can be directly added together, as opposed to arithmetic returns which require a more complex method of combination. Secondly, the original investment amount can be more readily back-calculated when logarithmic returns are employed.

2.2.2 Risk measurement

Following the same approach of using historical data, the variance of an asset is calculated as follows:

$$\text{Var}(r) = \sigma^2 = \frac{1}{m-1} \sum_{t=1}^m (r_t - \bar{r})^2 \quad (3)$$

where σ^2 is the portfolio variance;
 r_t is the portfolio return at time t ;
 $t = 1, 2, \dots, T$, where T is the number of time periods;
 $\bar{r}$ is the average asset return across all observations;
 m is the number of observations or time periods.

After determining the variance, we obtain the standard deviation, also known as volatility, which Markowitz's theory suggests as a key indicator of an asset's risk. This statistical metric is commonly utilized to gauge risk when analysing financial datasets. In essence, investors tend to seek assets with lower standard deviation, as it indicates less risk (see Equation 4).

$$\sigma(r) = \sqrt{\text{Var}(r)} = \sqrt{\frac{1}{m-1} \sum_{t=1}^m (r_t - \bar{r})^2} \quad (4)$$

where $\sigma(r)$ is the standard deviation of the portfolio returns.

The diversification level of the portfolio largely depends on the covariances between the securities that comprise the portfolio. Similar to how variance and standard deviation are derived, historical price data is employed to estimate the covariance matrix $\Omega_{(m \times n)}$ as follows:

$$\Omega_{(m \times n)} = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \cdots & \sigma_{1,n} \\ \sigma_{2,1} & \sigma_2^2 & \cdots & \sigma_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{m,1} & \sigma_{m,2} & \cdots & \sigma_m^2 \end{bmatrix} \quad (5)$$

where:

$$\sigma_{ij} = \text{cov}(r_{i,t}, r_{j,t}) = \frac{\sum_{t=1}^n (r_{it} - \bar{r}_i)(r_{jt} - \bar{r}_j)}{m} \quad (6)$$

where σ_i^2 is the variance of the returns for asset i ;
 σ_{ij} represents the covariance between the returns of asset i and asset j ;
 r_{it} and r_{jt} are the returns of assets i and j at time t , respectively;
 $\bar{r}_i$ and $\bar{r}_j$ are the average returns of assets i and j across all m periods;
 m is the number of observations or periods considered for each asset;
 i and j index assets within the portfolio, where $i, j = 1, 2, \dots, N$.

The matrix's diagonal entries signify the variances of the individual assets, as indicated by the covariance matrix.

2.2.3 Mean-Variance Model

Mean-Variance model, also known as Markowitz model, represents a quadratic optimization problem to either maximize portfolio returns for a defined level of risk, or conversely, to minimize risk, as measured by portfolio variance, for a set level of expected return. Let's detail the mathematical formulas for calculating a portfolio's expected return and its standard deviation. The portfolio's expected return is determined by taking the weighted average of the expected returns of each asset in the portfolio:

$$r_p = \sum_{i=1}^N x_i r_i \quad (7)$$

where r_p is the expected portfolio return;
 x_i is the proportion of the $i - th$ security in the portfolio;
 r_i is the expected return of the $i - th$ security;
 N is the number of securities in the portfolio.

The Markowitz portfolio theory also extends to managing uncertainty, framing the portfolio risk as a function of the diversity and correlation of the investments it contains. In this optimization framework, risk is quantified by the portfolio's variance, a measure of the combined volatility of the assets. The variance itself is derived from the covariance matrix of the returns of the assets in the portfolio, reflecting how each asset's returns move in relation to the others. The standard deviation of the portfolio, then, captures both the individual volatilities and the interaction of the asset returns ensuring that diversification.

The calculation of portfolio risk is presented in formula:

$$\sigma_p = \sqrt{\sum_{i=1}^N \sum_{j=1}^N x_i x_j \sigma_{ij}} \quad (8)$$

where x_i is the proportion of the $i - th$ security in the portfolio;
 x_j is the proportion of the $j - th$ security in the portfolio;
 σ_{ij} is the covariance of returns of the $i - th$ and $j - th$ securities;
 i and j index assets within the portfolio, where $i, j = 1, 2, \dots, N$.

This leads to the following quadratic optimization problem:

$$\begin{aligned} \text{Min } \sigma_p &= \sqrt{\sum_{i=1}^N \sum_{j=1}^N x_i x_j \sigma_{ij}} \\ \text{s.t. } &\sum_{i=1}^N x_i r_i \geq r_t \end{aligned} \quad (9)$$

$$\sum_{i=1}^N x_i = 1$$

$$x_i \geq 0$$

where r_t is the target return;

i and j index assets within the portfolio, where $i, j = 1, 2, \dots, N$.

Intuitively, the total of the asset weights must sum to one.

2.2.4 Mean-Semivariance Model

The downside risk model, was first conceptualized by A.D. Roy in 1952 (Roy, 1952). Around the same time, Harry Markowitz was developing his mean-variance theory. The model is rather an extension of the Markowitz theory than a completely new theory designed to solve some of the contradictions of the classical Markowitz model (Rom & Ferguson, 1994).

The main limitations of the Markowitz model are that it implies the following:

1. The standard deviation of the returns of the portfolio securities is a true measure of risk.
2. The returns of portfolio securities can be adequately represented by a normal distribution.

Modern portfolio theory uses measures of risk and return that do not always accurately reflect objective reality. Thus, using a normal distribution implies that the uncertainty of obtaining income above the expected (or average, using historical returns) is equal to the uncertainty of obtaining income below the expected. Also, the Markowitz model evaluates securities with yields in which the number of positive deviations from the average exceeds the number of negative ones as riskier than they actually are. The reverse distortion is also characteristic of yields, where negative yields prevail. H Markowitz himself noted that in some cases the SPT does not accurately reflect behavior and assumed that a model using half-dispersion would be more preferable, but due to the complexity of calculations of half-dispersion, he nevertheless took the mean and standard deviation as a basis (Sharpe, 1964).

Also, this formula introduces the concept of "target yield" for comparison with the actual yield, which also more accurately reflects the objective reality. This theory proposes a measure of risk known as "downside risk". An example of the calculation of this risk measure is shown in the formula:

$$\text{Semideviation} = \sigma_- = \sqrt{\frac{1}{m} \sum_{i=1}^m \min(r_i - r_{\text{target}}, 0)^2} \quad (10)$$

where σ_- is the semideviation, representing the "downside risk";

m is the number of returns;

r_i is the $i - th$ return;

i index assets within the portfolio, where $i = 1, 2, \dots, N$.

r_{target} is the target or required return;

$\min(r_i - r_{\text{target}}, 0)$ is the minimum operator ensuring only negative deviations are considered.

This formula accounts for the squares of returns below the target return. Squaring exponentially "penalizes" "bad" returns, which corresponds more closely to an intuitive understanding of risk (Fishburn, 1977).

However, "Academics and practitioners optimize portfolios using the mean-variance approach far more often than the mean-semivariance approach, despite the fact that semivariance is often considered a more plausible measure of risk than variance. The popularity of the mean-variance approach follows in part from the fact that mean-variance problems have well-known closed-form solutions, whereas mean-semivariance optimal portfolios cannot be determined without resorting to obscure numerical algorithms. This follows from the fact that, unlike the exogenous covariance matrix, the semicovariance matrix is endogenous, that is, a change in weights affects the periods in which the portfolio underperforms the benchmark, which in turn affects the elements of the semicovariance matrix.

In order to overcome this obstacle, many algorithms have been proposed to solve the problem:

- 1) Hogan and Warren (1972) propose to solve this problem using the Frank-Wolfe algorithm.
- 2) Nawrocki and Staples (1989) expand the scope of Nawrocki (1983) by considering the lower partial moment (LPM) as a risk measure.
- 3) Markowitz et al. (1993) transform the mean-semivariance problem into a quadratic problem by adding fictitious securities.
- 4) Ballesterio (2005), in turn, proposes a definition of portfolio semivariance (restricting the benchmark to the mean) that, when incorporated into optimization problems, these can be solved by applying parametric quadratic programming methods" (Estrada, 2015).

Nevertheless, I decided to proceed with heuristic approach, that yields a symmetric and exogenous semicovariance matrix, which enables the determination of mean-semivariance optimal portfolios by using the well-known closed-form solutions of mean-variance problems and allows to solve this problem without having any black-box numerical algorithm. As the results of the Javier Estrada (2015) study show, the heuristic approach portfolio semivariances are very highly correlated (around 0.98), as well as close in value (has difference under 1%), to the exact portfolio semivariances they intend to approximate.

The objective function and constraints of the heuristic Mean-Semivariance Model can be expressed as follows:

$$\text{Min } \sigma_{-p}^2 = \sum_{i=1}^N \sum_{j=1}^N x_i x_j \text{cosem}_{ij} \quad (11)$$

$$\text{s.t. } \sum_{i=1}^N x_i r_i \geq r_t$$

$$\sum_{i=1}^N x_i = 1$$

$$x_i \geq 0$$

where

$$\text{cosem}_{ij} = \frac{1}{T} \sum_{t=1}^T [\min(r_{it} - r_{\text{target}}, 0) \cdot \min(r_{jt} - r_{\text{target}}, 0)] \quad (12)$$

where r_{it} and r_{jt} are the returns for assets i and j at time t ;

r_{target} is the target or required return;

T is the number of observations;

i and j index assets within the portfolio, where $i, j = 1, 2, \dots, N$.

2.2.4.1 Performance analysis

Visually demonstrating the outcomes of both models can be effectively done by using a risk-return graph. This graph (see Figure 2) places the portfolio's risk, measured by standard deviation, on the horizontal axis, and the expected portfolio return on the vertical axis. On this plane, we can map out all potential portfolios. The efficient frontier is then identified as the curve where each point represents a portfolio offering the maximum expected return for a given amount of risk, or alternatively, the minimum risk for a given expected return (Ivanova & Kamenov, 2017).

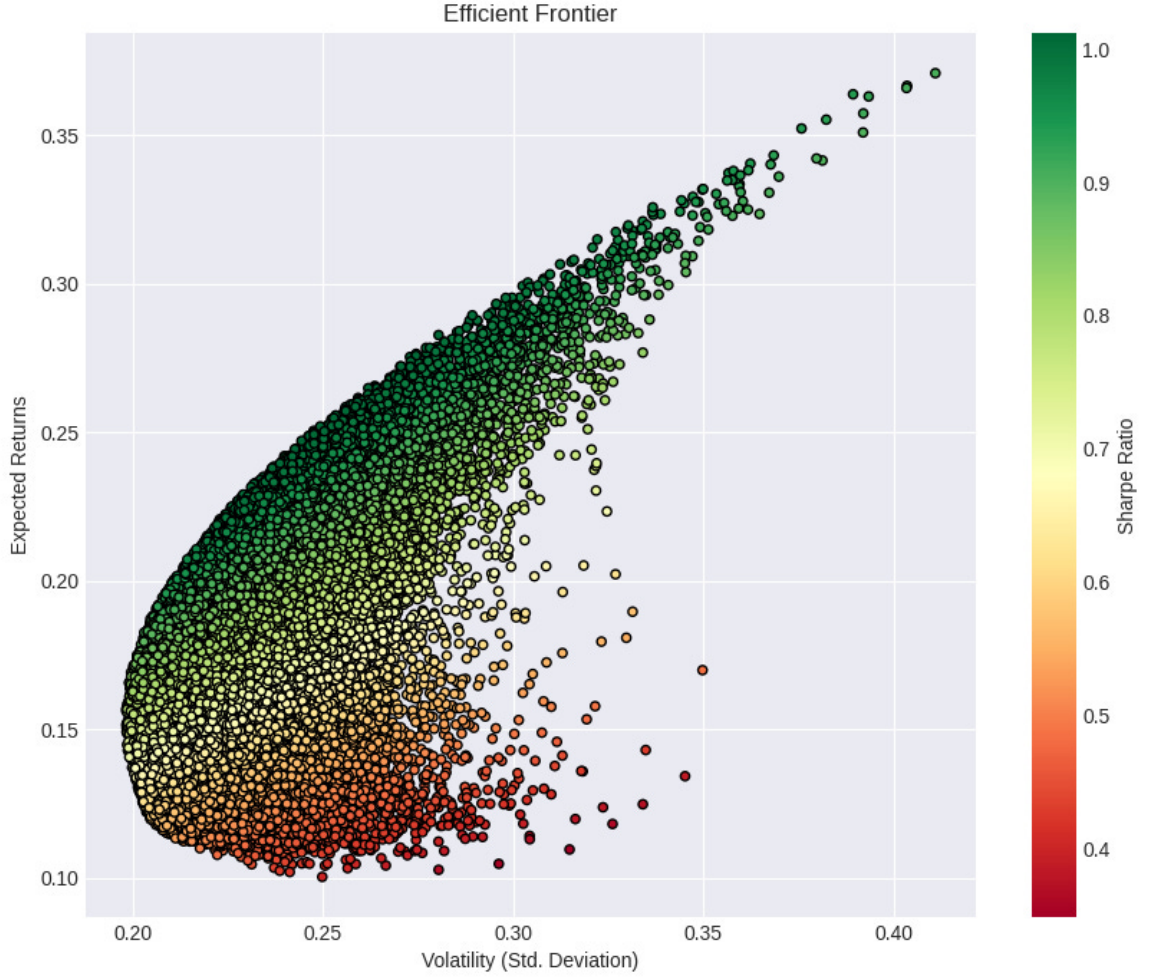


Figure 2 Efficient Frontier (Investments, 2022)

But how to get the optimal portfolio from the proposed set? The method of portfolio selection was proposed by W. Sharpe in 1966 and consists of ranking portfolios according to the ratio of return to risk, known as the Sharpe ratio (Sharpe, 1966).

The calculation of this coefficient is represented by the formula:

$$SR = \frac{R_p - R_f}{\sigma_p} \quad (13)$$

where R_p is the return of the portfolio;

R_f is the risk-free rate;

σ_p is the standard deviation of the portfolio return.

This formula is understood as follows - the calculated coefficient shows how many units of risky return are equal to 1 unit of risk. Accordingly, an investor should choose a portfolio with the highest Sharpe ratio.

As noted above, the Sharpe ratio, as well as the entire Mean-Variance model, is based on the assumption of a normal distribution of returns. Accordingly, when considering returns that do not meet the criterion of normal distribution, but, for example, are skewed to the right

(positive "surprises" are present, which the investor will assess as desirable), the Sharpe ratio will assess them as high risk.

To rank portfolios, the Mean-Semivariance theory suggests a coefficient similar to the Sharpe ratio - the Sortino ratio (Sortino & Price, 1994):

$$SR = \frac{R_p - R_f}{\sigma_-} \quad (14)$$

where R_p is the return of the portfolio;
 R_f is the risk-free rate;
 σ_- is the semideviation, representing the “downside risk”.

Also, one of the types of performance appraisal is the Relative Performance Analysis (RPA). It is a methodological approach employed to assess the performance of an investment model relative to a benchmark without relying solely on traditional risk-adjusted metrics such as Sharpe and Sortino ratios, which may not be feasible in all scenarios.

To navigate the limitations, Relative Performance Analysis focuses instead on direct comparisons of returns and more tangible metrics like drawdowns between the portfolio and the benchmark. This approach not only provides a clearer picture of performance over specified periods but also highlights resilience and risk exposure through maximum drawdown analysis. Such analysis is crucial for illustrating how effectively a portfolio strategy can outperform the benchmark while managing downside risk, offering a more nuanced understanding of portfolio performance in practical investment contexts.

The drawdown D_t at time t is calculated as the difference between the peak value before time t and the value at time t , expressed as a percentage of the peak value. This is mathematically represented as:

$$D_t = \frac{P_{\text{peak}} - P_t}{P_{\text{peak}}} \quad (15)$$

where P_{peak} is the highest value the investment has reached before time t ;
 P_t is the value of the investment at time t ;
 D_t is the drawdown;
 $t = 1, 2, \dots, T$, where T is the number of time periods.

2.3 Python

Python programming language, which due to its simplicity and flexibility has gained wide popularity in academia, data science and automation, is chosen to write the program and manipulate the data (Beazley, 2013).

The Python programming language is a high-level language and, therefore, supports all mathematical operations - operations of integral and differential calculus, statistical analysis, linear algebra, etc. However, this feature is also a disadvantage of this language, because all operations are performed in the Python virtual machine, where files are recompiled into byte-code and executed. But for highly loaded applications, it is extremely important to maximize

performance. To achieve the highest performance, mathematical calculations will be performed using third-party libraries.

2.3.1 NumPy

NumPy is an open source library for Python, is an implementation of the BLAS and LAPACK specifications for linear algebra problems, and supports JIT compilation to machine code. It is optimised for handling multidimensional arrays and supports high-level mathematical functions designed to handle such arrays. The part of the library responsible for data processing is written in compilable C, which significantly increases the speed of computation.

2.3.2 Pandas

Pandas is a Python programming library for data processing and analysis that acts as a kind of high-level API for the NumPy library, which is a lower-level tool. Pandas operates on special data structures called "Data Frames", which consist of data series, including time series, and provides the developer with a large set of operations from the field of statistical analysis, implemented as callable functions, which allows to abstract the mathematical apparatus and focus on data analysis.

2.3.3 Matplotlib

Matplotlib is a library in the Python programming language for visualizing data with 2D and 3D graphics. The resulting images can be saved to any image file supported by the operating system. Matplotlib is a flexible, easily configurable package that, together with NumPy, SciPy and IPython, provides capabilities like "MATLAB".

2.3.4 SciPy

SciPy is a fundamental library for scientific computing in Python, building on the capabilities provided by NumPy. It extends NumPy's capabilities by adding a collection of algorithms and high-level commands for manipulating and visualizing data. SciPy includes modules for optimization, linear algebra, integration, interpolation, and more, covering a wide range of mathematical operations. The library harnesses the power of compiled and optimized code, primarily written in C and Fortran, making it highly efficient for numerical computations. Its integration with NumPy arrays and its easy-to-use interface make it an indispensable tool for those conducting complex mathematical calculations and developing scientific applications.

3 Program development

In the applied section of our study, we begin by gathering data of publicly listed companies that meet our selection standards for constructing a portfolio. Subsequently, we implement two optimization approaches previously detailed in the theoretical discussion: the Markowitz Mean-Variance Model and the Mean-Semivariance Model, utilizing daily stock return data. For each approach, we outline the computational steps and chart the resulting efficient frontiers to visualize the optimal portfolios. We proceed to contrast the outcomes of these models with the performance of the Moscow Exchange index. The computational analysis and visualization in this section were performed using the Jupiter Notebook environment within the Python programming language.

3.1 Data

In order to apply our models we first need to prepare and clean the data to receive accurate results.

3.1.1 Tickers

The initial step in our process is selecting the stocks that will be used to construct our models. To this end, we've established multiple criteria. A key requirement is that the stock must be listed on the first quotation list of the Moscow Exchange, which maintains three such lists, each with its own set of inclusion criteria. To qualify for the first listing level, a stock must fulfill certain conditions:

- For companies with a market capitalization exceeding RUB 60 billion, there must be a minimum of 10% of shares in free float, which includes both ordinary and preference shares.
- Companies valued below RUB 60 billion follow a specific formula for free float, ensuring the total value of freely traded ordinary shares is at least RUB 3 billion and that of preference shares is RUB 1 billion.
- The issuer must have been registered for a minimum of three years.
- Financial statements prepared in accordance with IFRS must be available and disclosed for at least the last three full fiscal years.

The list of tickers of the shares of the first quotation list was obtained from the Moscow Exchange website (MOEX, 2024).

Additionally, it's important to highlight that some companies are dual-listed on the first quotation list due to having both ordinary and preferred shares, such as SBER and SBERP. The key distinctions between preference and ordinary shares include:

- Preference shares come with a fixed minimum dividend, whereas there is no obligation for ordinary shares to distribute dividends.
- Ordinary shares grant voting rights at shareholder meetings, while preferred shares offer voting rights only in certain exceptional situations.
- In the event of a company's liquidation, preferred share owners have priority over common share owners when it comes to payment.

Considering the focus of this paper is on individual investors, for companies with both common and preferred shares available, I have opted to include only preferred shares in the analysis. This is often due to their slight price discount relative to common shares.

3.1.2 Extracting the data

Since 2022, numerous international data services, including Yahoo Finance and Investing, ceased providing data on Russian stocks. The freely available Russian sources, like the Moscow Exchange, do offer closing prices but these are unadjusted for dividend disbursements and historical stock splits. Accurate modeling, however, necessitates the use of adjusted closing prices.

The data for this study was sourced from EOD Historical Data (see Code 2), a financial data provider that covers over 70 stock exchanges globally, the Moscow Exchange being one among them. This provider ensured access to the adjusted closing prices essential for the precision of our model computations (EOD, 2024).

Code 1 Importing the libraries (source: own)

```
from datetime import date
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import requests
from scipy import optimize
```

Code 2 Extracting the data (source: own)

```
russian_stocks = pd.read_excel("/Users/daniltimosevskij/Python/Russian
stocks.xlsx", sheet_name = 'List',)["ticker with P"]
russian_stocks = russian_stocks.dropna()
russian_stocks = russian_stocks.astype(str) + ".MCX"

api_key = '659c220d967ba1.28407049'
assets = russian_stocks
start_date = '2014-01-01'
end_date = '2023-12-31'
pf_data = pd.DataFrame()

for t in assets:
    url =
f'https://eodhistoricaldata.com/api/eod/{t}?from={start_date}&to={end_date}&api_t
oken={api_key}&fmt=json'
    response = requests.get(url)

    # Check the response status code
    if response.status_code != 200:
        print(f"Failed to get data for {t}: Status code {response.status_code}")
        continue
```

```

try:
    data = response.json()
    df = pd.DataFrame(data)
    df = df.set_index('date')
    pf_data[t] = df['adjusted_close']

except ValueError as e:
    print(f"Failed to parse JSON for {t}: {e}")

```

Data for Russian stocks were gathered for a ten-year stretch, beginning in early 2014 and concluding at the end of 2023. This dataset was subsequently structured into a DataFrame to facilitate easier manipulation and examination in subsequent stages. Overall, information was secured for a selection of 64 companies.

3.1.3 Cleaning the data

Upon reviewing the dataset, it became apparent that there were gaps in the data for certain stocks, where the adjusted closing price was missing, showing up as zeroes in the dataset.

date	AQUA.MCX
2014-02-10	27,8087
2014-02-11	0
2014-02-12	27,1825
2014-02-13	26,7705
2014-02-14	27,1825
2014-02-17	26,3667
2014-02-18	26,5315
2014-02-19	26,3667
2014-02-20	25,5922
2014-02-21	0
2014-02-24	25,5922

Table 2 Example of gaps for AQUA stock (source: own)

To prevent skewed return calculations, the decision was made to fill these gaps with the data from the previous cell with value (see Code 3).

Code 3 Cleaning the gaps (source: own)

```

for t in assets:
    pf_data[t] = pf_data[t].replace(0, np.nan) # Replace 0 with NaN
    pf_data[t] = pf_data[t].fillna(method='ffill') # Forward fill to replace NaN
with the previous non-NaN value

```

The final step before applying the models was to exclude any companies that have been trading for less than 4 years from our list (see Code 4). The rationale behind this decision is the belief that such stocks possess insufficient historical price data to accurately evaluate their performance.

Code 4 Excluding companies that have been trading for less than 4 years (source: author)


```

# Count non-NaN entries for each column
non_nan_counts = pf_data.count()

# Filter columns where the count is less than the threshold (e.g., 1000 for
roughly 4 years)
cols_to_keep = non_nan_counts[non_nan_counts >= 1000].index
pf_data = pf_data[cols_to_keep]
pf_data.columns

```

After all cleansing we have 43 issuers in the list meeting our conditions

3.2 Mean-Variance Model

We implemented the Mean-Variance Markowitz model on our dataset. Although this model typically presumes that returns are normally distributed, for the purposes of simplification, we have disregarded the non-normal nature of the stocks' distributions.

3.2.1 Logarithmic Returns

The first step in modelling we need to calculate the mean daily returns of all stocks, using logarithmic approach (see Code 5).

Code 5 Calculating the logarithmic daily return (source: own)

```

log_returns = np.log(pf_data/pf_data.shift(1))
mean_log_returns = log_returns.mean()* 250

```

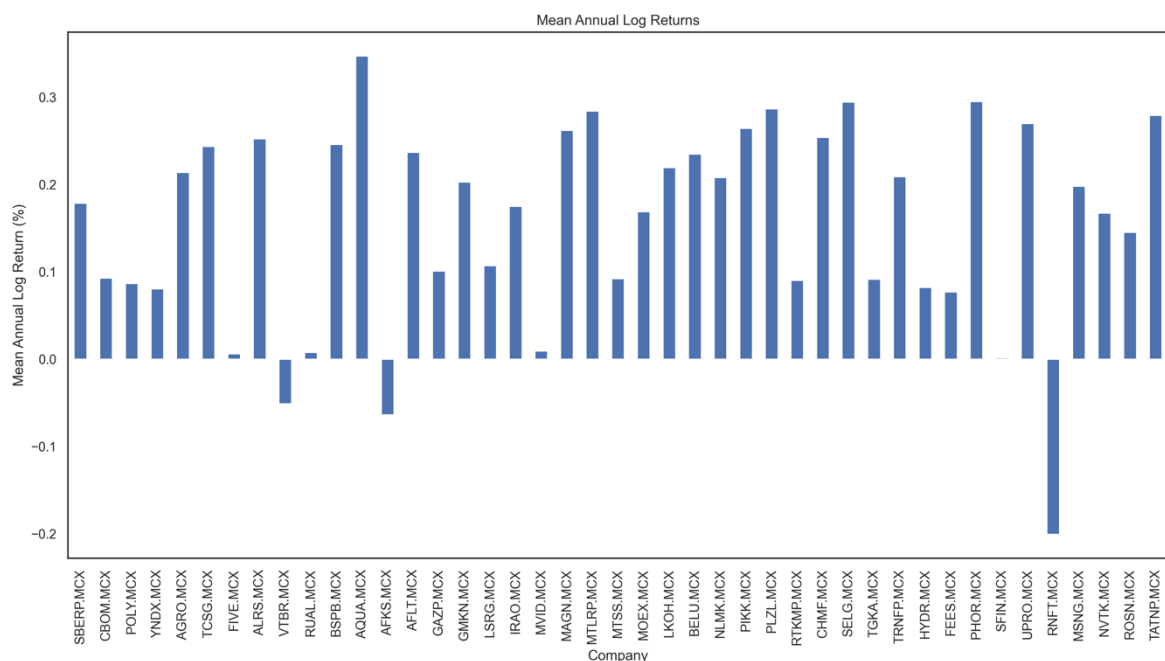


Figure 3 Mean Annual Log Returns (source: own)

The bar chart (see Figure 3) displays the mean annual log returns for all Russian stocks which we explore. The returns vary across different companies, with most showing positive mean annual log returns between 10-30% per year, while a few, such as VTBR and AFKS, depict negative returns, suggesting a decline over the period analyzed. Only one particular company RNFT stands out with a significantly lower return around -20% compared to others, indicating a substantial decrease in value.

3.2.2 Covariance and Correlation

Having calculated the returns of all stocks for all periods, we can calculate the covariance matrix and the correlation matrix (see Code 6).

Code 6 Calculating the covariance and correlation matrices (source: own)

```
cov_matrix = log_returns.cov() * 250
correlation_matrix = cov_matrix.corr()
```

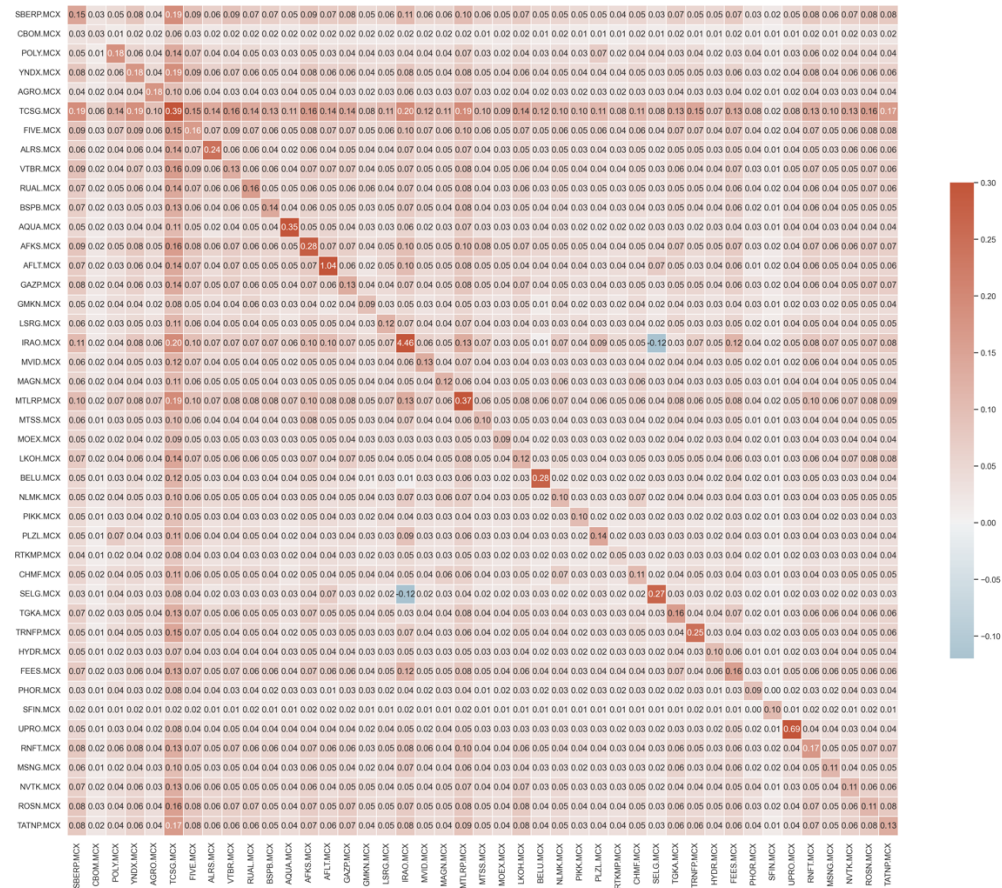


Figure 4 Covariance Matrix for Mean-Variance model (source: own)

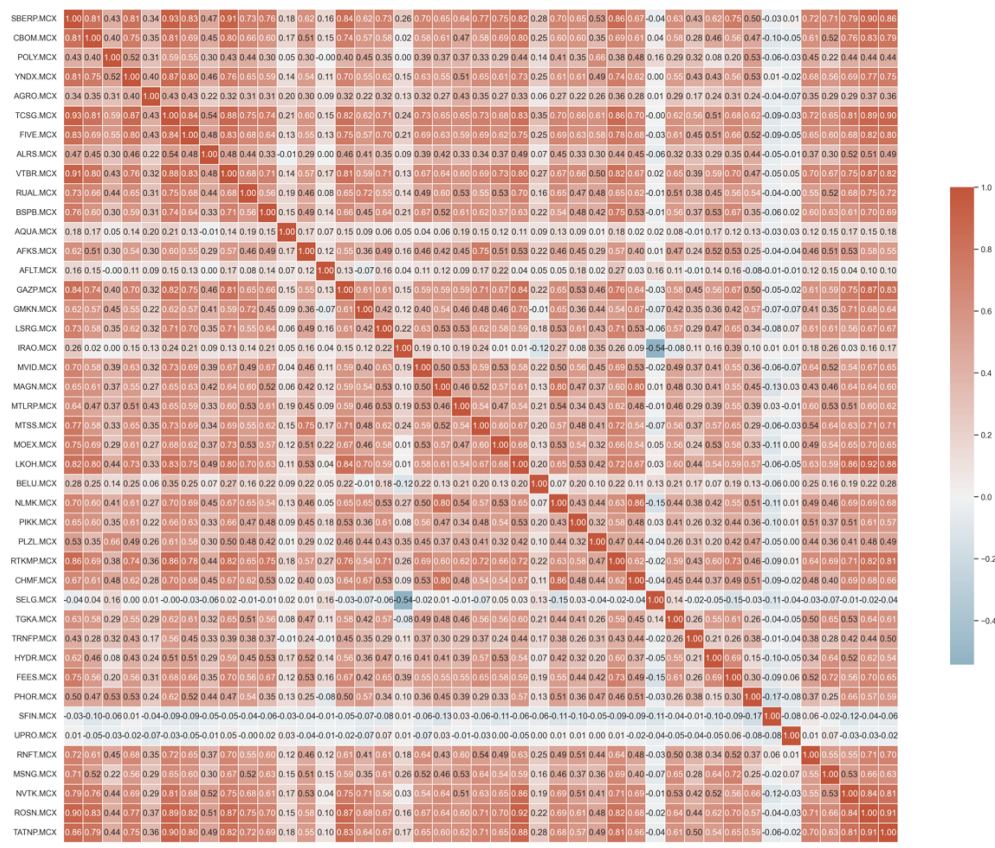


Figure 5 Correlation Matrix for Mean-Variance model (source: own)

From the correlation matrix (see Figure 5), we see that the returns of most stocks are in a linear relationship and the intensity of correlation is mostly moderate to high positive, as indicated by the predominance of the predominance of warm colours. There are only a few instances, such as AQUA, AFLT, IRAO, SELG, SFIN and UPRO of low or negative correlation, suggested by the neutral or cool colours.

3.2.3 Mean-Variance model calculation

In this section, we apply the Markowitz model to our dataset. As outlined in Chapter 2, the core of the Markowitz model is to minimize the variance of portfolio returns, which is essentially trying to get the best possible return for the lowest risk. To tackle this problem, we've employed the SciPy library (see Code 7).

The code sets up the optimization process where it adjusts the investment weights across different stocks to find the combination that gives us the highest Sharpe Ratio. This is done by the SciPy optimization function, which is cleverly told to minimize the negative Sharpe Ratio – it's a mathematical trick because the function is designed to find minimum values, and we want the maximum Sharpe Ratio.

We've also set some rules for the optimization: the total investment must add up to 100% (no borrowing or shorting), and we can't put a negative amount into any stock. The method we've chosen from SciPy's toolkit is 'SLSQP', which stands for Sequential Least

Squares Programming. It's a method well-suited for the kind of constraints we're dealing with, like making sure we don't end up with a negative amount of any stock or investing more than we have.

We've also set some rules for the optimization: the total investment must add up to 100% (no borrowing or shorting), we can't put a negative amount into any stock, and the maximum allocation for any stock in the portfolio is 10%. As we know that the application of the model is not a 100% guarantee that the resulting portfolio will be effective and in the future, so that it was decided to limit the weight of one company to 10% in order to maintain diversification. The method we've chosen from SciPy's toolkit is 'SLSQP', which stands for Sequential Least Squares Programming. It's a method well-suited for the kind of constraints we're dealing with, like making sure we don't end up with a negative amount of any stock or investing more than we have.

Code 7 Calculating the Mean-Variance model with the highest Sharp ratio (source: own)

```
# Some helper functions
def get_ret_vol_sr(weights):
    weights = np.array(weights)
    ret = np.sum(weights * log_returns.mean()) * 250
    vol = np.sqrt(np.dot(weights.T, np.dot(log_returns.cov() * 250, weights)))
    sr = (ret - risk_free_rate) / vol
    return np.array([ret, vol, sr])

# Negate Sharpe ratio as we need to max it but Scipy minimizes the given function
def neg_sr(weights):
    return get_ret_vol_sr(weights)[-1] * -1

# Constraints for the optimization problem
def check_sum(weights):
    return np.sum(weights) - 1

cons = {'type': 'eq', 'fun': check_sum}

# bounds on weights
bounds = tuple((0, 0.1) for _ in range(num_assets))

# Initial guess for optimization to start with
init_guess = [(100 / num_assets * 0.01) for _ in range(num_assets)]

# Call minimizer for highest SR
opt_results = optimize.minimize(neg_sr, init_guess, constraints=cons,
                                bounds=bounds, method='SLSQP')

optimal_weights = opt_results.x
```

After determining the optimal weights, I refine the distribution of our investment portfolio based on the outputs obtained from the Markowitz Mean-Variance model optimization. Upon inspection of the initial weight allocations, it was observed that certain assets were assigned marginal weights of less than 1% of the portfolio. From a strategic

investment perspective, such negligible allocations are operationally inefficient and contribute little to the portfolio's performance.

Consequently, we implemented a reallocation algorithm (see Code 8). This algorithm zeroes out the weights of all assets falling below the 1% threshold, thereby consolidating the investment into assets with more substantial weights. The liberated capital from the excluded assets is then proportionally redistributed among the remaining assets that originally had weights between 1% and 10%, ensuring no individual asset allocation exceeds the 10% cap.

This reallocation is performed iteratively to adjust for any rounding discrepancies that may occur during the redistribution process, thereby maintaining the precision of the portfolio construction. The final step in this algorithm re-normalizes the portfolio weights, ensuring the sum of the weights is equal to unity, which upholds the fundamental constraint of the model that the collective investment proportions must encompass the entirety of the portfolio's capital.

This refined approach aligns with prudent portfolio management practices, ensuring that the investment is not only optimized for risk-adjusted returns but is also practically implementable without being diluted by excessively fractional allocations.

Code 8 Refining the distribution of the investment portfolio (source: own)

```
def adjust_portfolio_weights(opt_weights, cap=0.1):
    # Round down any weights below 1% to 0
    opt_weights[opt_weights < 0.01] = 0

    # Normalize the weights to sum to 1 after rounding down
    opt_weights /= opt_weights.sum()
    # Identify the indexes of the stocks that are at the cap
    capped_indexes = opt_weights >= cap

    # Cap the weights at the cap for stocks above the cap
    excess_weight = (opt_weights[capped_indexes] - cap).sum()
    opt_weights[capped_indexes] = cap

    # Identify the indexes of stocks with weights less than the cap
    distributable_indexes = opt_weights < cap

    # Distribute excess weight to stocks with weights below the cap
    # proportionally
    opt_weights[distributable_indexes] += (excess_weight *
                                           (opt_weights[distributable_indexes] /
                                            opt_weights[distributable_indexes].sum()))

    # Ensure weights sum up to 1 after redistribution
    opt_weights /= opt_weights.sum()

    return opt_weights
```

The ● symbol identifies the portfolio with the highest Sharpe ratio on the efficient frontier (see Figure 6), indicating an optimal balance between return, which stands at 0.277, and deviation, at 0.1972. For the purposes of this evaluation, we have simplified the

computation by assuming a 0% risk-free asset return when determining the portfolio. The resulting Sharpe ratio is 1.404, suggesting that this portfolio is the most efficient in terms of risk-adjusted return. It's also noteworthy that our adjusted risk-free return overstates the Sharpe ratio.

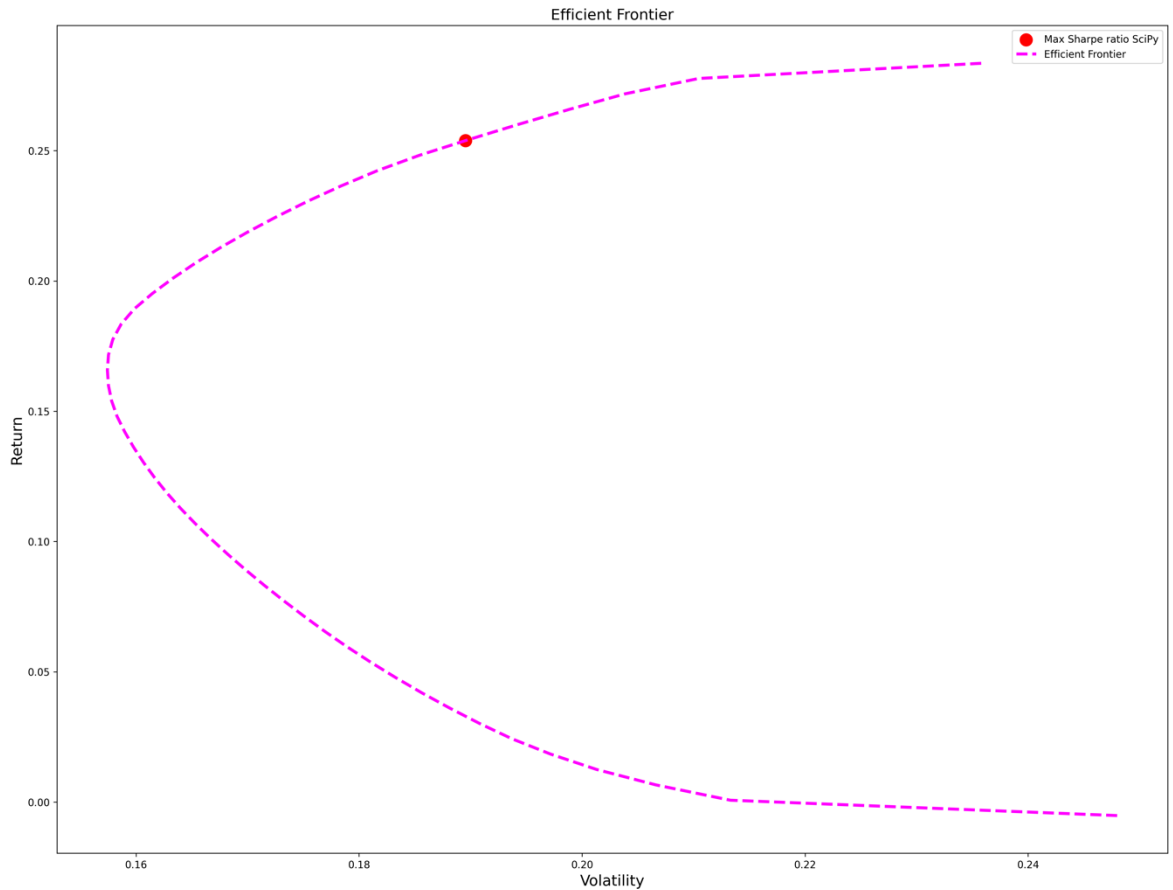


Figure 6 Efficient frontier for Mean-Variance model (source: own)

The obtained results of the ● portfolio weights can be seen in the Figure 7 below:

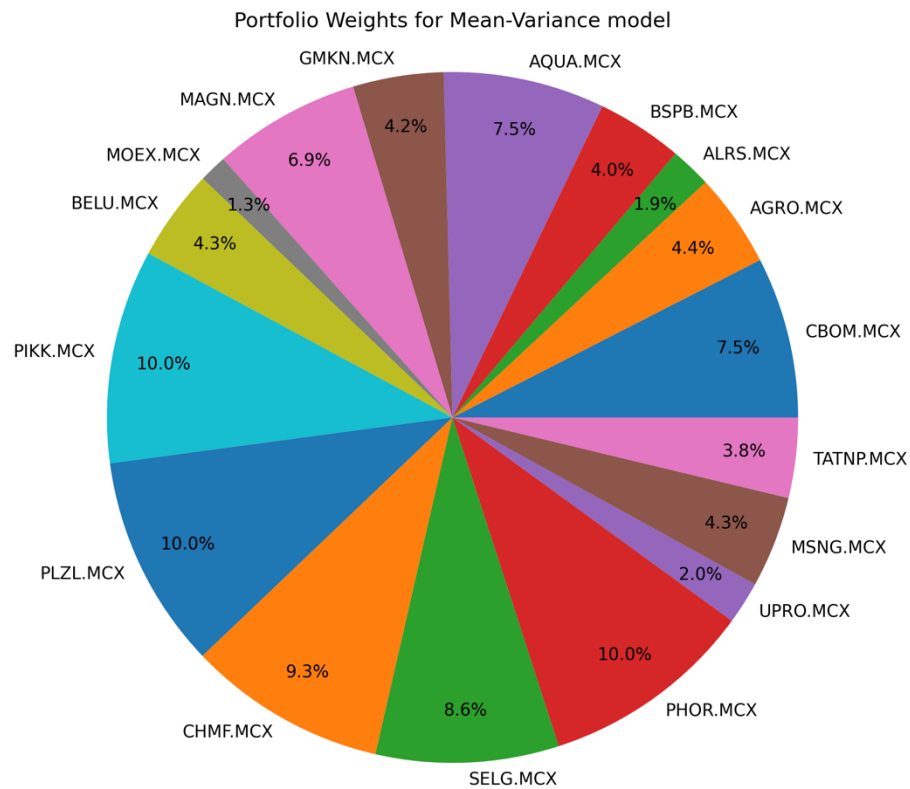


Figure 7 Optimal portfolio for Mean-Variance model (source: own)

3.3 Mean-Semivariance Model

In this subsection, we apply the Mean-Semivariance model to the data. As mentioned earlier the main difference of Mean-Semivariance model is the approach to risk assessment, the so-called downside risk.

3.3.1 Downside risk

Choosing the right method for calculating semi-deviation, where the main difference between the different approaches is the choice of the target or required return, can lead to different portfolio constructions. I would distinguish 3 main approaches:

1. **Stock Values Below Zero.** This method measures the volatility of negative returns only, focusing purely on losses. It's straightforward and easy to compute, but It may not provide a realistic measure of risk for all types of investments, especially for assets that rarely dip below zero in terms of returns (e.g., indexes).
2. **Stock Values Less Than the Fixed return.** The term Fixed return most often refers to either the desired targeted return or the market average. This approach is more dynamic than measuring against a zero return, allowing for a better understanding of how an asset performs relative to a typical market environment. However, the static parameter might not accurately reflect changing market conditions. May not be sensitive enough to recent market trends or anomalies.

3. Stock Values Less Than the Market Return (Dynamic Benchmark). Given method is most responsive to market conditions, using a dynamic benchmark that adjusts with market performance. It allows for a nuanced view of risk by comparing to how the market is performing over the same period, rather than a long-term average.

Taking into consideration all 3 possible approaches I decided to proceed with calculation of semicovariance matrix using dynamic benchmark since it is the most accurate in terms of volatile market conditions.

3.3.2 Covariance and Correlation

Already having calculated logarithmic stock returns before we use them to calculate Semicovariance focusing only on underperformance and setting all positive differences to 0 (see Code 9).

Code 9 Calculating the semicovariance and correlation matrices (source: own)

```
def calculate_semi_covariance_matrix(returns, benchmark):  
  
    # Calculate differences from the benchmark, focus on underperformance  
    differences = returns.sub(benchmark, axis=0)  
    # Set all positive differences to 0, focusing only on the negative  
    differences = differences.clip(upper=0)  
    # Calculate the semi-covariance matrix  
    semi_cov_matrix = differences.cov() * 250 # Annualize  
  
    return semi_cov_matrix  
  
semi_cov_matrix = calculate_semi_covariance_matrix(log_returns, in_log_returns)  
correlation_matrix_ms = semi_cov_matrix.corr()
```

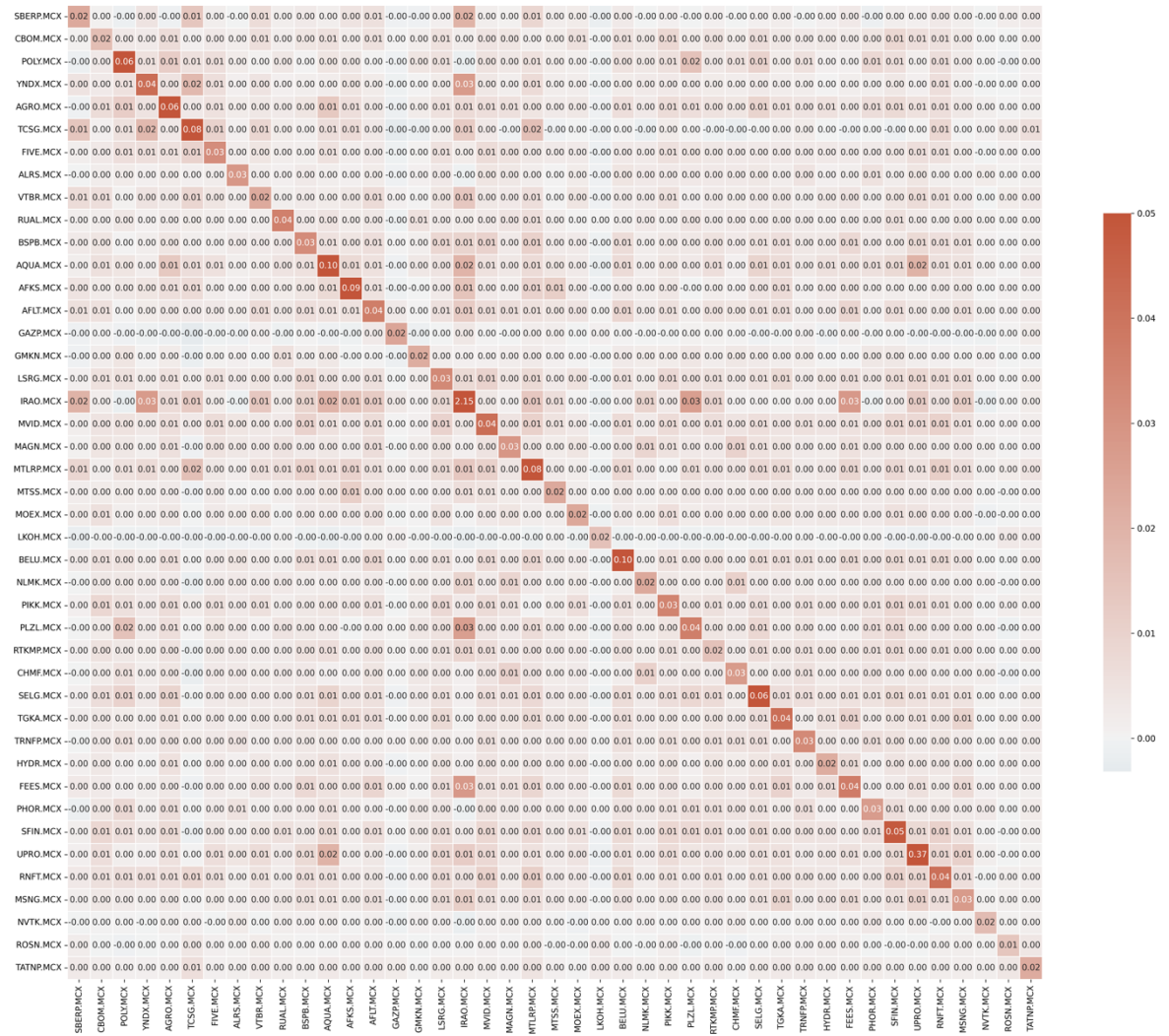



Figure 8 Semicovariance Matrix for Mean-Semivariance model (source: own)

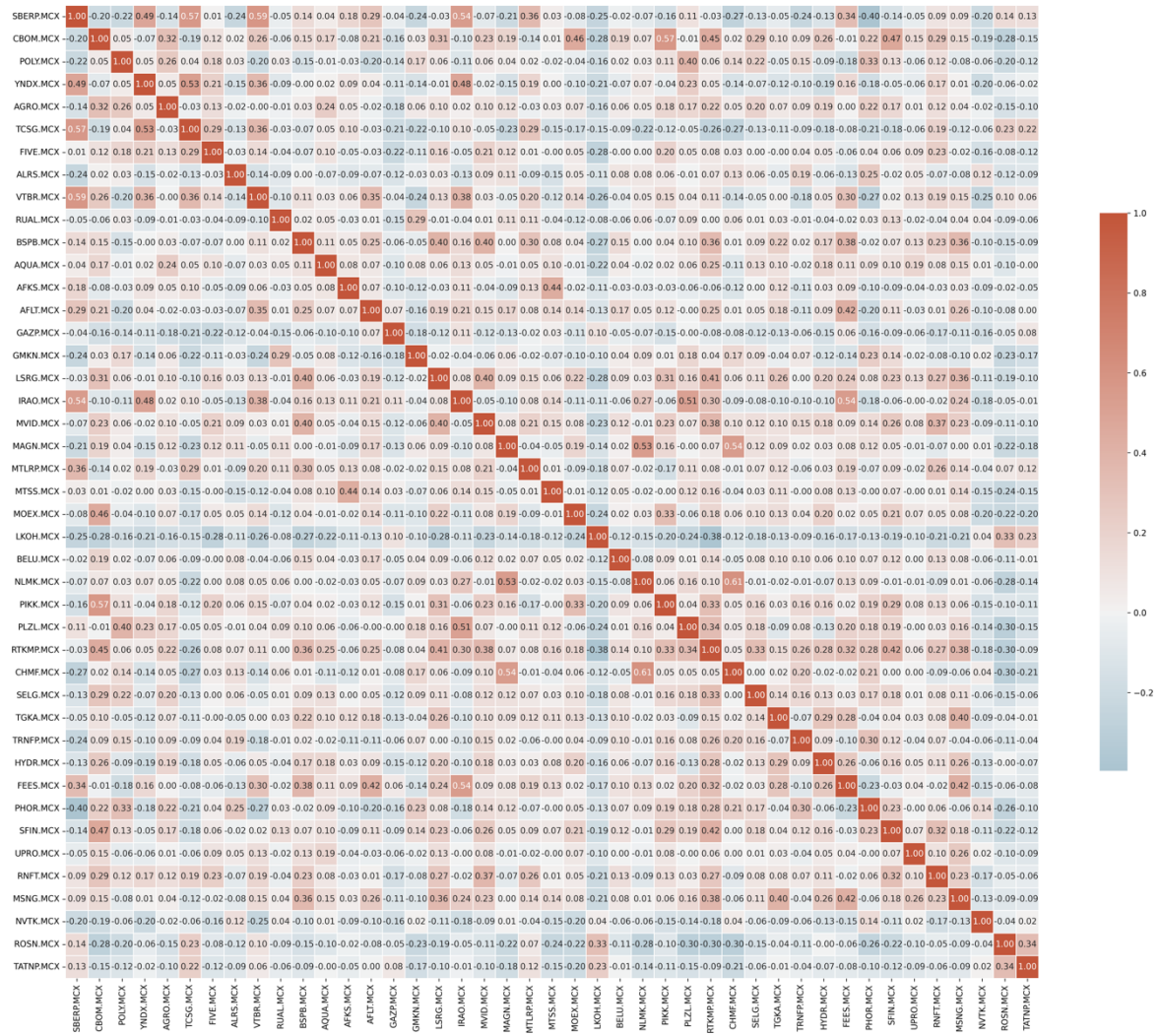


Figure 9 Correlation Matrix for Mean-Semivariance model (source: own)

3.3.3 Mean-Semivariance model calculation.

Similar to the approach described in Chapter 3.2.3, we use the SciPy library to implement the optimization process. However, instead of minimizing variance, we minimize the portfolio's semivariance - a proxy for downside risk (see Code 10).

I retain the same constraints as the mean-variance model: the sum of the weights equals 100%, no negative weights are allowed, and we limit the maximum allocation for any single stock to 10%. These constraints help maintain portfolio diversification and manage risk by preventing overexposure to any single asset. Instead of Sharpe ratio is used Sortino ratio as the performance metric.

The optimization is performed with SciPy's 'SLSQP' algorithm, which is designed for problems with linear and non-linear constraints. The process seeks the set of weights that maximizes this Sortino ratio, thus aiming for the portfolio with the highest expected return per unit of downside risk.

Code 10 Calculating the Mean-Semivariance model with the highest Sortino ratio (source: own)

```
def get_ret_semidev_sr(weights):
```

```

weights = np.array(weights)
ret = np.sum(weights * log_returns.mean())* 250
portfolio_semivar = np.dot(weights.T, np.dot(semi_cov_matrix, weights))
semidev = np.sqrt(portfolio_semivar)
sr = (ret - risk_free_rate) / semidev
return np.array([ret,semidev,sr])

# Negate Sharpe ratio as we need to max it but Scipy minimizes the given function
def neg_sr_semidev(weights):
    return get_ret_semidev_sr(weights)[-1] * -1

# Negate the return as we need to max it but Scipy minimizes the given function
def highest_return_semidev(weights):
    return get_ret_semidev_sr(weights)[0] * -1

# Constraints for the optimization problem
def check_sum(weights):
    return np.sum(weights) - 1

cons = {'type':'eq', 'fun':check_sum}

# bounds on weights
bounds = tuple((0, 0.1) for _ in range(num_assets))

# Initial guess for optimization to start with
init_guess = [(100/num_assets * 0.01) for _ in range(num_assets)]

# Call minimizer for highest SR
opt_results_semidev = optimize.minimize(neg_sr_semidev, init_guess,
constraints=cons, bounds=bounds, method='SLSQP')

```

We also adjust the obtained results using the previously created function `adjust_portfolio_weights()` and output the effective curve with the optimal portfolio designation ● (see Figure 10).

Code 11 Adjusting the optimal weights (source: own)

```

optimal_weights_semidev = opt_results_semidev.x
optimal_weights_semidev = adjust_portfolio_weights(optimal_weights_semidev)

```

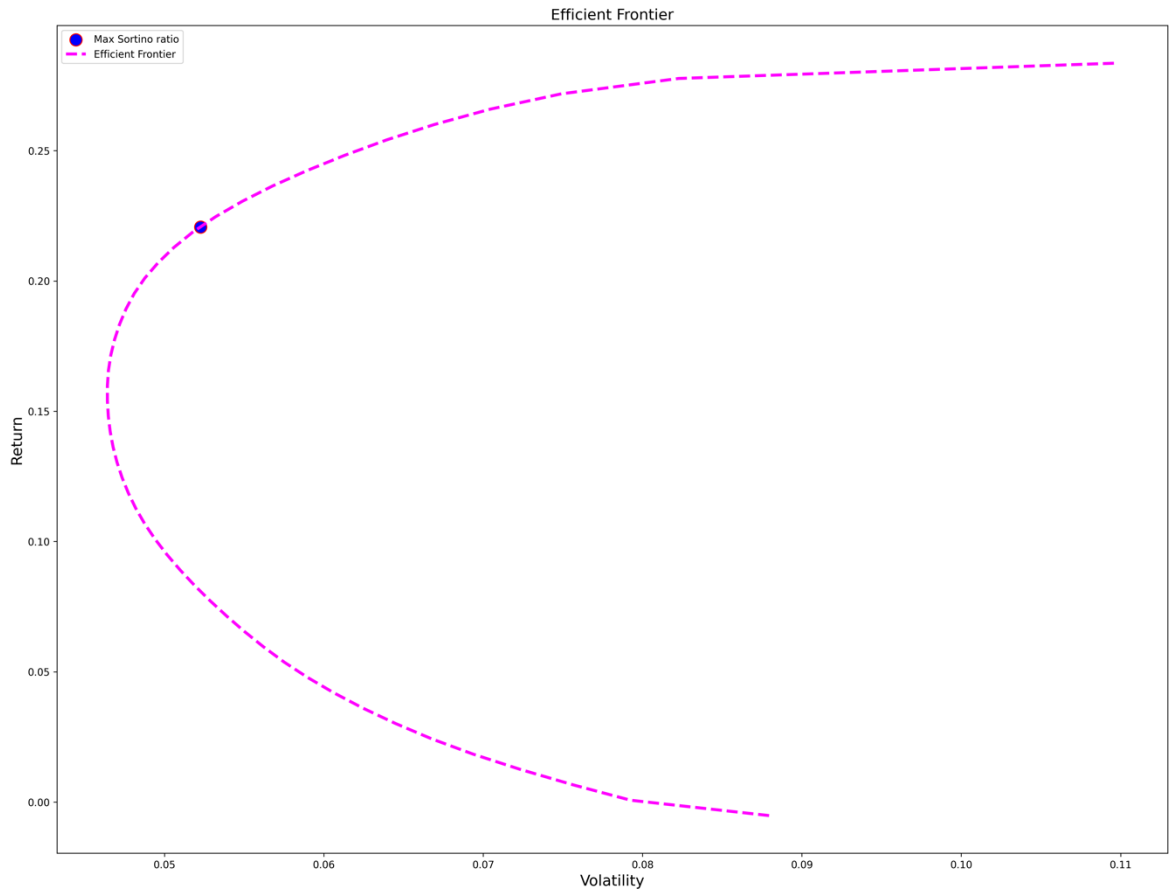


Figure 10 Efficient frontier for Mean-Semivariance model (source: own)

Specifically, this portfolio delivers a return of 0.2207, with a measured downside volatility of 0.0523, mainly caused by small deviations from the dynamic index. By incorporating a 0% risk-free asset return for the purpose of this analysis, we attain a Sortino Ratio of 4.2215. This indicates an exceptionally efficient portfolio when assessing risk-adjusted return with a focus on downside risk. The weights in the portfolio are represented in the Figure 11 below.

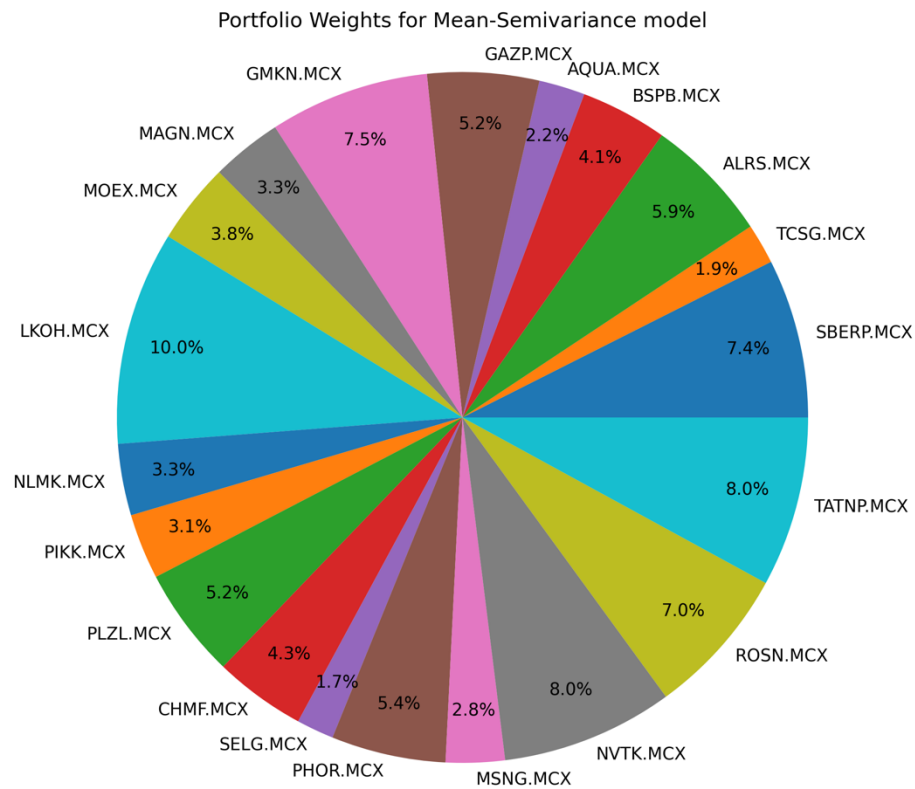


Figure 11 Optimal portfolio for Mean-Semivariance model (source: own)

3.4 Comparison of results

In this part of the thesis, we will compare the obtained results. In order to make the evaluation process more efficient, it was decided to compare the resulting portfolios not only with each other but also with the Moscow Exchange index.

3.4.1 Comparison of two models

The Table 4 presented below outlines the allocation weights of various stocks within the optimal portfolios derived from both the Mean-Variance (MV) and Mean-Semivariance (MS) models. The data reveals a notable divergence in the portfolio compositions, which is attributed to the models' distinct approaches to risk evaluation. It is apparent that a direct performance comparison of the models may not be entirely valid due to their different risk assessment methodologies.

Nevertheless, if we provide a comparison of relative returns and use standard deviation as their common measure of risk (see Table 3). In this context, the Mean-Variance model has achieved a historical return of 25.4%, while the Mean-Semivariance model has yielded a return of 22.0%. Furthermore, the comparative analysis detailed in the table shows the Mean-Variance model exhibiting not only a higher return but also a lower volatility at 19.0% compared to the Mean-Semivariance model's 23.3%. This difference in volatility underpins a notable distinction in risk profiles between the two models and leads to two different Sharpe ratios of 1.339 and 0.947 respectively.

While using the semideviation as the measure of volatility, the Mean-Variance model displays a semideviation of 0.085, indicating more significant negative fluctuations compared to the Mean-Semivariance model's lower semideviation of 0.052. This finding suggests that the Mean-Semivariance model, despite its higher overall volatility (0.246 standard deviation versus 0.190 for MV), manages downside risks more effectively.

The implications of these measurements on the Sortino ratios further elucidate the models' performances. The Mean-Variance model, with a Sortino ratio of 2.99, indicates less favorable performance in managing downside risk compared to the Mean-Semivariance model, which achieves a higher Sortino ratio of 4.22.

In conclusion, the choice between the Mean-Variance and Mean-Semivariance models depends on the investor's risk tolerance and specific focus on either total volatility or downside risk alone. While the Mean-Variance model offers a balanced approach with lower overall volatility, it exhibits higher potential for negative fluctuations as reflected in its semideviation. In contrast, the Mean-Semivariance model, with its lower semideviation and higher Sortino ratio, could be preferable for investors prioritizing effective downside risk management and seeking potentially higher risk-adjusted returns in more turbulent market conditions.

I personally favor the Mean-Semivariance model for several compelling reasons, particularly in volatile markets like Russia. This model's reliance on semideviation as a measure of risk provides a more nuanced understanding of downside risk, which is crucial in unstable or bearish markets. Moreover, the use of a dynamic benchmark in calculating the Mean-Semivariance model aligns more closely with fluctuating market conditions, making it highly responsive and potentially more robust in anticipating and reacting to market downturns. Given these considerations, the Mean-Semivariance model is particularly advantageous for investors who prioritize downside protection and are operating in volatile environments.

Portfolio	Return	Std. deviation	Sharp ratio	Semideviation	Sortino ratio
MV	0,254	0,190	1,339	0,085	2,99
MS	0,137	0,246	0,55	0,052	4,22

Table 3 Mean-Variance and Mean-Semivariance comparison (source: own)

Stock	MV Weight (%)	MS Weight (%)
SBERP.MCX	0	7,44
CBOM.MCX	7,53	0
POLY.MCX	0	0
YNDX.MCX	0	0
AGRO.MCX	4,39	0
TCSG.MCX	0	1,89
FIVE.MCX	0	0
ALRS.MCX	1,91	5,89
VTBR.MCX	0	0
RUAL.MCX	0	0
BSPB.MCX	4,02	4,06
AQUA.MCX	7,55	2,15

AFKS.MCX	0	0
AFLT.MCX	0	0
GAZP.MCX	0	5,22
GMKN.MCX	4,24	7,48
LSRG.MCX	0	0
IRAO.MCX	0	0
MVID.MCX	0	0
MAGN.MCX	6,9	3,31
MTLRP.MCX	0	0
MTSS.MCX	0	0
MOEX.MCX	1,3	3,79
LKOH.MCX	0	10
BELU.MCX	4,28	0
NLMK.MCX	0	3,35
PIKK.MCX	10	3,07
PLZL.MCX	10	5,19
RTKMP.MCX	0	0
CHMF.MCX	9,29	4,29
SELG.MCX	8,56	1,75
TGKA.MCX	0	0
TRNFP.MCX	0	0
HYDR.MCX	0	0
FEES.MCX	0	0
PHOR.MCX	10	5,36
SFIN.MCX	0	0
UPRO.MCX	2	0
RNFT.MCX	0	0
MSNG.MCX	4,27	2,76
NVTK.MCX	0	8,05
ROSN.MCX	0	7,02
TATNP.MCX	3,75	7,96

Table 4 Mean-Variance and Mean-Semivariance weights (source: own)

3.4.2 Comparison of Mean-Variance model with MOEX

In the analysis, we extend the application of portfolio assessment to encompass a comparison with the Moscow Exchange index (MOEX). This necessitates the procurement of index data corresponding temporally to the dataset used for the Markowitz model, enabling the computation of historical returns and volatility for the derivation of the Sharpe ratio (see Code 12). Employing this ratio as a critical comparative metric, I will evaluate the performance of the investment portfolios. The assessment aims to determine the relative efficiency of the Mean-Variance model against the benchmark index, thus providing an empirical basis for the examination of portfolio optimization strategies.

Code 12 Calculating the Return, Deviation and Sharpe ration for Moscow exchange index (source: author)


```

index = 'MCFTR.INDX'
in_data = pd.DataFrame()

url =
f'https://eodhistoricaldata.com/api/eod/{index}?from={start_date}&to={end_date}&a
pi_token={api_key}&fmt=json'
response = requests.get(url)

data = response.json()
df = pd.DataFrame(data)
df = df.set_index('date')
in_data = df['adjusted_close']

in_data = in_data.replace(0, np.nan) # Replace 0 with NaN
in_data = in_data.fillna(method='ffill')

annualized_in_std = in_log_returns.std() * np.sqrt(250)
risk_free_rate = 0
average_annual_in_return = in_log_returns.mean() * 250
sharpe_ratio_in = (average_annual_in_return - risk_free_rate) / annualized_in_std

```

The Table 5 below presents the results from our portfolio performance comparison. It includes the returns, volatility, and Sharpe ratio for the Mean-Variance (MV) portfolio alongside the Moscow Exchange index. The Mean-Variance portfolio shows a return of 25.4% and volatility of 19.0%, resulting in a Sharpe ratio of 1.339. On the other hand, the Moscow Exchange index has a return of 13.7%, a higher volatility of 24.6%, and a Sharpe ratio of 0.55.

The comparison shows that the Mean-Variance portfolio has a better performance than the benchmark index, as indicated by a higher Sharpe ratio, which suggests a better risk-adjusted return. These findings support the effectiveness of the Mean-Variance model in optimizing a portfolio compared to the performance of the broader market, highlighting its value for informed asset allocation.

Portfolio	Return	Volatility	Sharp ratio
MV	0,254	0,190	1,339
Index	0,137	0,246	0,55

Table 5 Comparison of Mean-Variance model with index (source: author)

3.4.3 Comparison of Mean-Semivariance model with MOEX

A direct comparison between the Mean-Semivariance model and a benchmark index utilizing traditional risk-adjusted return metrics, such as the Sortino ratio, is not feasible. The inherent limitation arises from the utilization of the index as a dynamic benchmark within the Mean-Semivariance model, which seeks to minimize deviations below the benchmark itself. Consequently, the semideviation of the index, relative to its own performance, is null, and the Sortino ratio for the index is either undefined or, theoretically, infinite. This makes a direct comparison using traditional risk-adjusted return metrics like the Sortino ratio problematic.

In order to compare the performances I decided to conduct Relative Performance Analysis (see Code 13), where I directly compare the returns and analyze the drawdowns of the portfolio and the index (see Figure 12). As already have been indicated before the average return of the index from 2014 to 2024 was 13.7% while the optimal portfolio Mean-Semivariance model shows 22.1%. If we compare the drawdowns over the same time intervals, the chart below shows that the optimal portfolio of Mean-Semivariance model performs better over the entire horizon and furthermore the maximum drawdown of the Mean-Semivariance portfolio was 46.2% while the index drawdown was 53.5%.

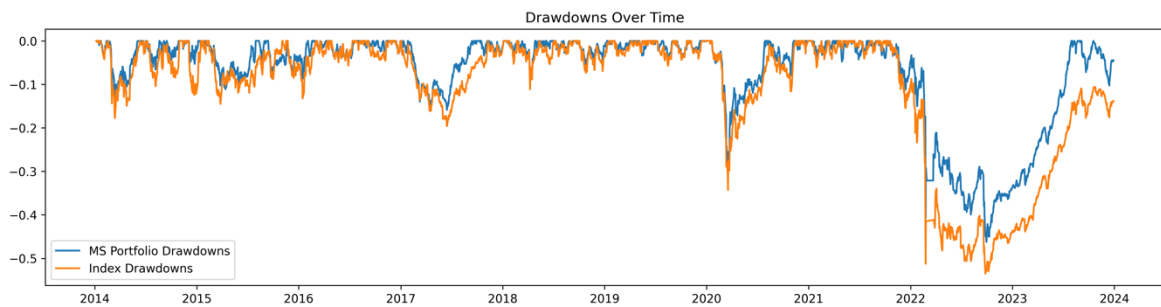


Figure 12 Drawdowns Over time for Mean-Semivariance and index (source: own)

Code 13 Calculating the drawdowns for Mean-Semivariance model and MOEX (source: own)

```
weighted_log_returns = log_returns.multiply(optimal_weights_semidev,
axis='columns')
semidev_pfolio_returns = weighted_log_returns.sum(axis=1)

def calculate_weighted_cumulative_returns(pf_data, weights, initial_prices):
    # Normalize prices by dividing by the initial prices
    normalized_prices = pf_data / initial_prices

    weighted_prices = normalized_prices.multiply(weights, axis='columns')

    combined_normalized_price = weighted_prices.sum(axis=1)

    # Calculate cumulative returns for the portfolio
    portfolio_cum_returns = combined_normalized_price
    return portfolio_cum_returns

def calculate_cumulative_returns(prices):
    # Normalize prices to start from the first price
    cumulative_returns = prices / prices.iloc[0]
    return cumulative_returns

def calculate_drawdowns(cumulative_returns):
    drawdowns = (cumulative_returns - cumulative_returns.cummax()) /
cumulative_returns.cummax()

    # Get the date of the maximum drawdown
    max_drawdown_date = drawdowns.idxmin()
    return drawdowns.min(), max_drawdown_date, drawdowns
```

```

weights = pd.Series(optimal_weights_semidev, index=pf_data.columns)
initial_prices = pf_data.iloc[0]

# Now calculate the portfolio cumulative returns
portfolio_cum_returns = calculate_weighted_cumulative_returns(pf_data, weights,
initial_prices)

# Calculate the index cumulative returns
index_cum_returns = calculate_cumulative_returns(in_data)

portfolio_max_drawdown, portfolio_max_dd_date, portfolio_drawdowns =
calculate_drawdowns(portfolio_cum_returns)
index_max_drawdown, index_max_dd_date, index_drawdowns =
calculate_drawdowns(index_cum_returns)

portfolio_cum_returns.index = pd.to_datetime(portfolio_cum_returns.index)
index_cum_returns.index = pd.to_datetime(index_cum_returns.index)

portfolio_drawdowns.index = pd.to_datetime(portfolio_drawdowns.index)
index_drawdowns.index = pd.to_datetime(index_drawdowns.index)

```

Conclusion

The main objective of this thesis was to develop a Python program capable of optimizing a share portfolio within the Russian stock market. This research concentrated on the examination and application of two principal models: the conventional Mean-Variance model, commonly referred to as the Markowitz model, and the Mean-Semivariance model, which employs a dynamic benchmark to establish the target return.

The practical deployment of these models necessitated an initial step of connection to API provided the Moscow exchange securities data, followed by the process of downloading and refining day-to-day stock data for all securities listed in the first-tier category, ensuring the removal of any undesirable data inconsistencies.

Subsequent to the data preparation phase, I engineered Python scripts to compute the optimal portfolios according to both models, incorporating predefined constraints, as author's conditions, concerning for the maximum and minimum stock weightings. A comprehensive comparison of the resultant portfolios was then conducted, contrasting them against the Moscow Exchange index benchmark, and the findings were systematically visualized.

The study concluded that, while it was determined that the resulting portfolios differ significantly from each other in terms of asset allocations due to a different approach to risk, both models demonstrated superior performance in comparison to the benchmark in terms of return and risk efficiency.

I also believe that the Mean-Semivariance model is particularly well-suited for the Russian market due to its emphasis on downside risk, which is crucial in a volatile environment such as Russia's and the use of the dynamic benchmark, effectively capturing the intricacies of market conditions.

It is imperative to acknowledge that these models, do not represent infallible solutions, rather, they offer alternative lenses through which to analyze market dynamics. The allure of deriving an optimal asset distribution from model-based data analyses is undeniable, yet, it is crucial to recognize the inherent limitations of quantitative models. One notable limitation is the reliance on historical data, which, as is widely known, may not be a reliable indicator of future performance. Additionally, focusing only on numbers can lead to the fact that broader contexts and relationships will be overlooked. Therefore, when making a real decision on how to allocate funds among possible investment opportunities, it is recommended that the investor use, in addition to quantitative analysis, also qualitative analysis.

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